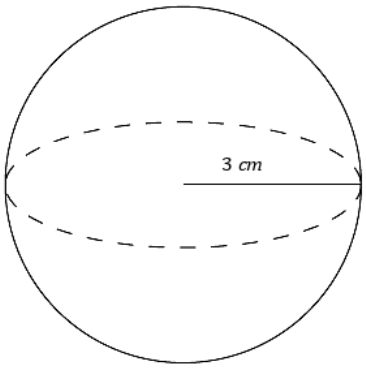
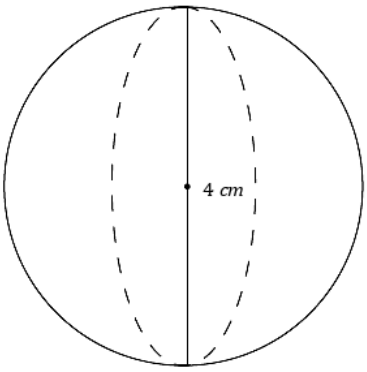
Volume of spheres and hemispheres

1 Find the volume of the following spheres, correct to two decimal places:

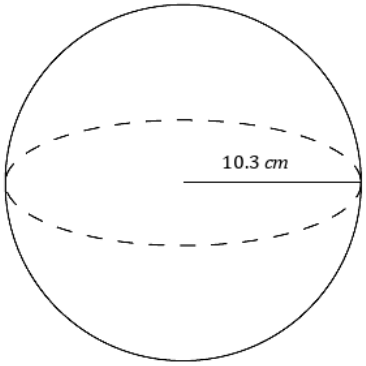
a



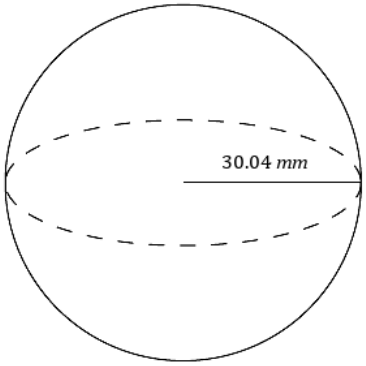
b



c



d



2 Find the radius of the following spheres, correct to two decimal places:

a A sphere with a volume of $167\ \text{cm}^3$.

b A sphere with a volume of $\dfrac{343\pi}{3}\ \text{cm}^3$.

3 The planet Mars has a radius of $3400\ \text{km}$.
Find the volume of Mars. Write your answer in scientific notation to three decimal places.

4 The planet Jupiter has a radius of $69\,911\ \text{km}$, and planet Mercury has a radius of $2440\ \text{km}$.
Assume that both planets are spheres.
How many times bigger is the volume of Jupiter than Mercury? Round your answer to one decimal place.

5 A ball has a volume of 904.779 cubic units.
Find the radius of the ball. Give your answer correct to the nearest tenth.

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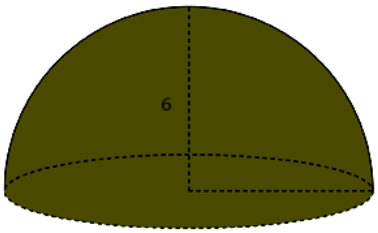
6 The radius of a bowling ball is $10.9\ \text{cm}$.
Find the volume of the ball. Round your answer to three decimal places.

7 The volume of a sphere is $20\,579.526\ \text{cm}^3$.
Find the diameter of the sphere. Give your answer correct to one decimal place.

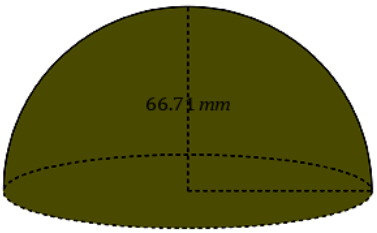
8 If the radius of a sphere doubles, by what factor will its volume increase?

9 Find the volume of the following solids, correct to two decimal places:

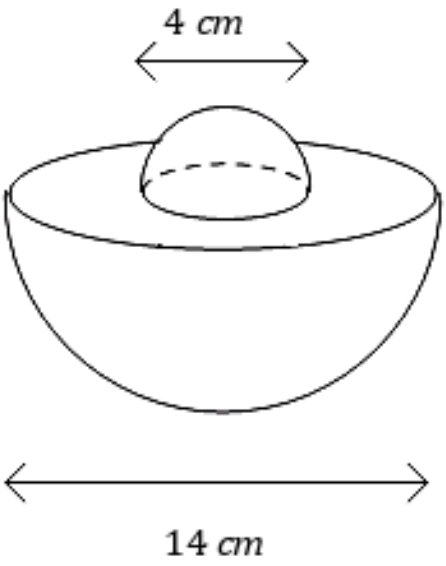
a



b



c

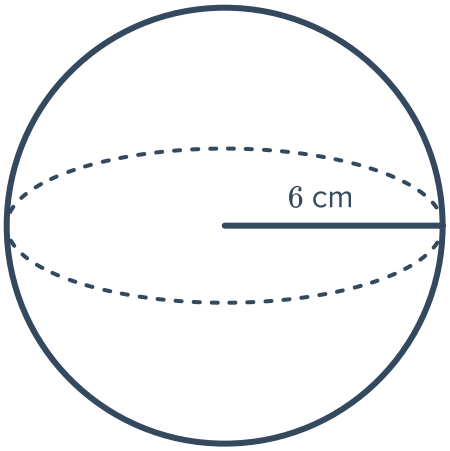


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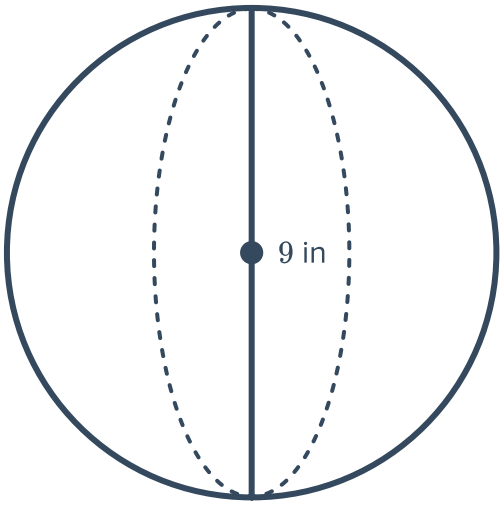
Additional questions

10 For each sphere, calculate the volume:

a



b



11 The diameter of a bowling ball is $8.6\ \text{in}$. What is the approximate volume of the ball?

12 The volume of a sphere is $972\pi\ \text{yd}^3$. What is the diameter of the sphere?

13 Tennis balls are packaged and sold in cylindrical containers, with three balls per container. The height of a standard tennis ball container is $21\ \text{cm}$.

a What is the volume of a tennis ball?

b What is the volume of the container?

c What percent of the container is occupied by the tennis balls?

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